

Multiple Choice

1. **Answer:** D

Options: A) a hand lens with 1 X magnification; B) a microscope with 40 X magnification; C) a light microscope with 100 X magnification; D) a dissecting microscope with 4 X magnification

Note: Correct option: D - a dissecting microscope with 4 X magnification

2. **Answer:** D

Options: A) deposition in water; B) slow cooling and crystallization; C) weathering and erosion; D) exposure to heat and high pressure

Note: Correct option: D - exposure to heat and high pressure

3. **Answer:** C

Options: A) 1; B) 2; C) 3; D) 4

Note: Correct option: C - 3

4. **Answer:** C

Options: A) Pressure has been converted to light.; B) Force has been converted to light.; C) Closing an electric circuit has produced light.; D) Opening an electric circuit has produced light.

Note: Correct option: C - Closing an electric circuit has produced light.

5. **Answer:** B

Options: A) acquired trait to gain an inherited trait.; B) inherited trait to gain an acquired trait.; C) acquired trait to gain another acquired trait.; D) inherited trait to gain another inherited trait.

Note: Correct option: B - inherited trait to gain an acquired trait.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'organically produced vegetable oil'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'ability to convert sunlight into energy'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It moves faster in the same direction it was moving.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** label. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

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End of Answer Key